

Volume 7: Admin Console, Security, DevOps & Deployment

Technical specification for platform administration, user moderation, sponsor provisioning, rate-limiting, secret encryption at rest, Docker infrastructure, and CI/CD pipelines.

Target Stack: NestJS 11 · Docker Compose · PostgreSQL 16 · Redis 7 · Nginx · Lead: Muhammad Sameer Ali

VOLUME OBJECTIVE:

Specifies the operational controls, user moderation tools, sponsor credential management, brute-force mitigation, containerized Docker architecture, and step-by-step production deployment checklist.

1. Admin Console & Platform Moderation

- **User Controls:** ``GET /admin/users``, ``POST /admin/users/:id/ban``, ``POST /admin/users/:id/balance-adjust``.
- **Sponsor Management:** ``POST /admin/sponsors`` generates sponsor accounts with reversibly-encrypted credentials (AES-256-GCM ``secret-box``).
- **Audit Logs:** All admin mutations are recorded immutably in ``AuditLog`` with IP address and timestamp.

2. Security Architecture & Threat Mitigation

Threat / Domain	Mitigation Strategy	Implementation Details
Brute-Force Login	Distributed IP & Account Throttle	AuthThrottle table; locks account for 15 mins after 5 fails.
Token Hijacking	Token Version Invalidation	User.tokenVersion checked on every request; revoked on logout-all.
Secret Exposure	Secret-Box Encryption	TOTP secrets & sponsor keys encrypted at rest via AES-256-GCM.
WebSocket Flooding	Turn Rate Limiter	Redis token-bucket enforces max 1 move per turn second per socket.

3. Production Infrastructure & Docker Topology

- **PostgreSQL 16:** Connection pool with Prisma Client (port 5432).
- **Redis 7:** In-memory cache, WebSocket pub/sub adapter, and BullMQ task queues (port 6379).
- **NestJS Backend:** Clustered Node.js server behind Nginx reverse proxy (port 4000).
- **Next.js Frontend:** Production build served on port 3000.

4. Backend Developer Deployment Checklist

- [] Run database migrations (``npx prisma migrate deploy``).
- [] Execute seed script (``npm run db:seed``) to create treasury, admin, demo players, and 7 campaign templates.
- [] Configure environment variables (Solana RPC, Redis URL, JWT Secret, Secret-Box Key).
- [] Start services via Docker Compose (``docker compose up -d``).
- [] Run E2E verification tests (``npm run test:e2e``).